

Tableau-driven Data Analysis and Visualization of COVID-19 Cases in INDIA

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Abstract—In this paper, we present a comprehensive analysis and visualization of COVID-19 data in India using Tableau, emphasizing key metrics such as the total number of cases, the distribution of confirmed cases by age group and gender, and the total number of deaths across states. Additionally, the paper delves into state-wise testing details, the count of testing labs in each state, and the administration of COVID-19 vaccines. The visualizations of this report dynamically show the pandemic's trajectory, accompanied by interactive graphs showing how many cases there are over time. Further insights are derived from demographic breakdowns, illustrating the proportion of confirmed cases by age group and gender. This analysis not only aids in understanding the impact on various demographic segments but also aids in making targeted public health interventions. A specific focus is placed on mortality, with state-wise graphs detailing the total number of deaths. These visualizations contribute to a nuanced understanding of the geographic distribution of the severity of the pandemic, aiding policymakers in resource allocation and response planning. The report also incorporates a detailed examination of testing strategies, providing state-wise testing details and highlighting the number of testing labs in each region. This information is crucial for assessing the robustness of testing infrastructure and guiding future testing strategies. The integration of Tableau in this analysis ensures that the visualizations are informative and interactive, allowing users to explore the data in real-time and customize views based on their specific interests or inquiries. By presenting this comprehensive analysis, the paper aims to contribute valuable insights to the collective understanding of the COVID-19 pandemic in India and provide a practical resource for decision-makers, researchers, and the general public.

Index Terms—COVID-19 impact, Tableau data analysis, Data visualization, Demographic patterns.

I. INTRODUCTION

The onset of the COVID-19 pandemic in India marked a significant and challenging period characterized by a substantial impact on public health, the economy, and societal norms. The nation witnessed a rapid surge in cases, leading to strain on healthcare facilities and a notable loss of lives. The stringent measures implemented, such as lockdowns, aimed to curb the virus's spread, albeit with consequential effects on businesses, employment, and vulnerable demographics. The pandemic has had a devastating impact on India, with over 35 million confirmed cases [1] and over 4 lakh deaths [2] reported as of November 6, 2023. This paper is devoted to shedding

light on the intricate dynamics of the pandemic within the context of India, employing a Tableau-driven approach to unravel the complexity of the public health crisis. As the nation grapples with the challenges presented by the virus, it becomes imperative to harness advanced data visualization and analytics tools to understand and respond to the evolving situation comprehensively. Against this backdrop, our study adopts a holistic approach to explore critical dimensions of the pandemic's impact. Beginning with examining the total number of cases, we delve into the demographic distributions to uncover patterns and variations that may inform targeted interventions. By leveraging the power of Tableau [3,4,5], our analysis transforms raw data into dynamic visualizations, offering stakeholders a real-time, interactive lens through which to navigate the intricate landscape of COVID-19 in India.

In addition to understanding the spread of the virus, our research extends to investigating state-wide mortality rates, providing crucial insights into the geographic variations in the severity of the pandemic. This understanding is essential for resource allocation, strategic planning, and policy formulation to address the healthcare challenges posed by COVID-19 effectively. Moreover, we delve into the testing landscape, examining the extent of testing efforts and collaboration details with the Indian Council of Medical Research (ICMR), a pivotal partnership in the nation's response to the pandemic. As the global scientific community works tirelessly to develop and deploy vaccines, our analysis extends to the vaccination campaign in India. We explore the number of doses administered and the broader context, including regional variations and the overall progress toward achieving widespread immunity. This paper serves as a gateway to a comprehensive analysis of the COVID-19 pandemic in India, utilizing Tableau to transform data into actionable insights. By exploring various dimensions, we aim to contribute to the ongoing discourse surrounding the pandemic, offering valuable perspectives for policymakers, healthcare professionals [6], and researchers engaged in the collective effort to mitigate the impact of COVID-19 on the Indian populace.

The rest of the paper is organized as follows. Section II presents the datasets used in the Tableau-driven analysis and visualization, and Section III explains the Tableau-driven data analysis and visualization using the COVID-19 dataset in India. Section IV shows the Tableau dashboard design

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using COVID-19 datasets. Finally, Section V concludes with Tableau-driven data analysis and visualization of COVID-19 cases in India.

II. DATASETS USED IN THE TABLEAU-DRIVEN ANALYSIS AND VISUALIZATION

Our Tableau-driven analysis and visualization use various data sources, including the Kaggle COVID-19 dataset in India, the Age group dataset, Individual details, and the ICMRT testing dataset. For more information on the datasets used in our analysis, refer to [7,8,9].

A. COVID-19 Dataset in India

This dataset includes statistics on the daily number of COVID-19 cases, fatalities, and recoveries in India. The information is accessible from January 30, 2020, until August 11, 2021. The dataset may be used to compare the effects of the virus on various states and union territories, evaluate the effectiveness of the government's reaction to the pandemic, study the virus's progress across India, and create prediction models to anticipate the pandemic's future trajectory.

B. State Wise Testing Details

This file contains state-wise daily details of the number of COVID-19 tests performed in India. The data spans from March 16, 2020, to October 24, 2021, providing a comprehensive overview of testing efforts during the peak period of the pandemic in India. It ensures a comprehensive view of the national testing by encompassing all states and union territories. Researchers and elected officials can follow the virus's path through various geographical areas by examining the number of daily tests and positive cases. Researchers can determine whether testing is sufficient to capture the accurate scale of the virus spread by comparing the number of tests performed to the size of the population.

C. COVID-19 Vaccine Statewise

This file contains daily state-wide details of COVID-19 vaccination in India. The data is available from January 16, 2021, to August 13, 2021. The dataset can be used to track the progress of the vaccination campaign, compare the vaccination rates in different states and union territories, and analyze the impact of various factors on vaccination coverage. The data also monitors the total doses administered, first and second-dose coverage, and vaccination rates across different regions. Analyzing vaccination rates across states and demographics can reveal disparities in access and uptake, informing targeted interventions. Researchers can use data analysis techniques to forecast future vaccination trends and plan resource allocation accordingly.

D. Age Group Details

The agegroupdetails.csv file contains information about the distribution of COVID-19 cases across different age groups in India. The data is collected from January 22, 2020, to May 31, 2021. This dataset helps to analyze the age-related patterns of COVID-19 cases in India and reveals susceptible

age groups. This allows us to look at the data to gauge the disease severity, rate of hospitalization, and mortality for each age category. Comparisons within the same data group before and after specific interventions, such as lockdowns, could also be done. Analyzing such data will enable the development of targeted messages and intervention programs for different age groups regarding their particular concerns.

E. ICMR Testing Details

This file contains information about the ICMR-approved testing laboratories for COVID-19 in India. The data includes the name of the laboratory, the city, the state, and the type of laboratory (government or private). The dataset can be used to identify the availability of COVID-19 testing facilities across different regions in India. Analyzing the location of laboratories can help identify areas with limited access to testing facilities and inform efforts to expand testing infrastructure. Analyzing the ratio of laboratories to population can help determine whether the existing testing infrastructure meets the population's needs.

F. Individual Details

The individualdetails.csv file contains individual-level data on COVID-19 cases in India. The data includes the patient's details like age, sex, history of travel, contacts with the confirmed cases, source of infection, state of residence, and outcome (dead, cured, or discharged). By assessing where people are getting infected with the virus, we shall know how the disease is spreading in different regions and target the necessary measures. By analyzing data as per age, location, and other similar factors, appropriate measures that are tailored for specific groups can be developed.

III. COVID-19 CASES IN INDIA: TABLEAU-DRIVEN DATA ANALYSIS AND VISUALISATION

The first set of visualizations outlines the trajectory of the pandemic, illustrating the total number of confirmed cases across the country. This dynamic presentation allows for a quick and intuitive grasp of the scale and progression of COVID-19 in the country. Fig. 1 shows the total number of confirmed cases and their status, whether they are cured or the patient died. The blue curve indicates the number of confirmed cases, while the orange and red curves show the number of cured and dead cases. The graph shows that the number of cases has been increasing significantly. There have been over 35 million confirmed cases of COVID-19 in India. From Fig. 1, it is clear that the scale of dead people is in thousands while that of confirmed and cured is in millions. This suggests that a large share of confirmed cases were cured.

Further, the analysis delves into the demographic impact of the virus. Visualizations showcase the proportion of confirmed cases distributed across different age groups, shedding light on potential vulnerabilities within specific demographics. The pie chart in Fig. 2 shows the age group distribution of COVID-19 cases in India. The pie chart shows that the working class population is the most affected age group. The age groups

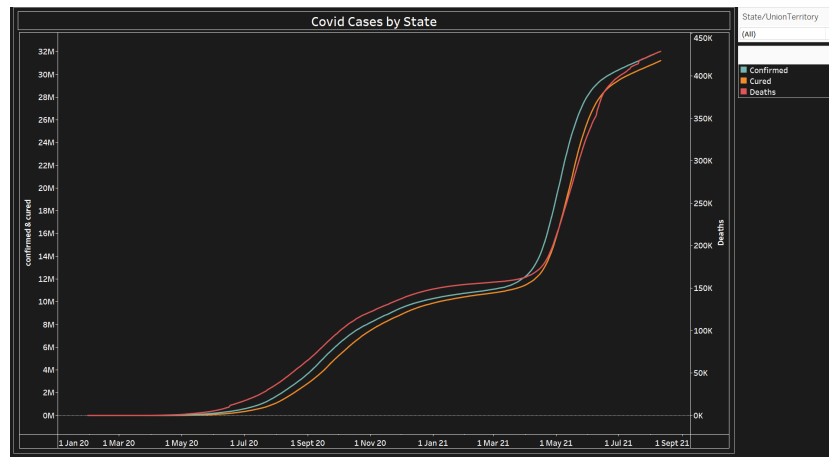


Fig. 1. Total number of cases and their status (confirmed, cured, and dead).

40-49, 30-39, and 20-29 account for 16.18%, 21.10%, and 24.86% of cases, respectively. The elderly are also significantly affected. The age group 60-69 accounts for 12.86% of all cases, followed by the age group 50-59 with 11.13% of cases. Children and adolescents are less affected but still at risk. The age group 10-19 accounts for only 3.90% of cases, while the age group 0-9 accounts for 1.30%. These observations suggest that COVID-19 is a serious threat to all age groups, but especially to older people and the working-age population. It is essential to continue to monitor the age distribution of COVID-19 cases to identify any changes in trends. This information can be used to inform public health policies and interventions.

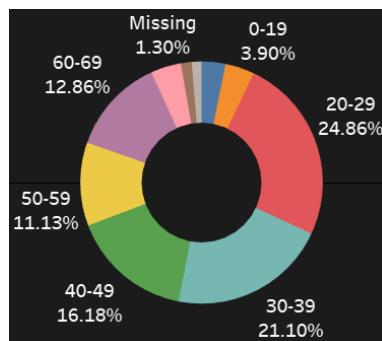


Fig. 2. Age group analysis.

Similarly, gender-based visualizations provide insights into the varying impact on different population segments. Fig. 3 shows the pie chart giving the gender distribution details of the individuals affected by the disease. The pie chart shows that 66.76% of the people affected by COVID-19 in India are male, while 33.24% are female. This means that men are nearly twice as likely to be affected by COVID-19 as women. There are a few possible explanations for this increased rate of COVID-19 cases in men. One possibility is that men are more likely to work in jobs with a higher risk of exposure to the virus, such as healthcare, transportation, and construction.

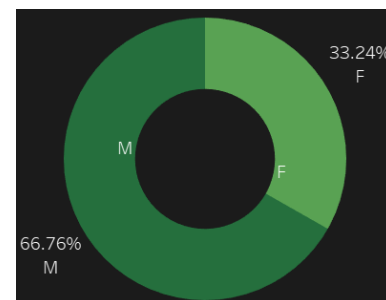


Fig. 3. Gender without missing values.

Another possibility is that men are more likely to engage in behaviors that increase their risk of exposure, such as smoking and drinking alcohol. Finally, it is also possible that biological factors, such as differences in immune system function, may play a role. It is important to note that this pie chart is based on data from a single study. More research is needed to confirm these findings and better understand why men are more likely to be affected by COVID-19. Analyzing this data can provide insights into whether there are any gender-specific trends or disparities in the contraction of the virus.

Mortality is a crucial focus of this analysis, with state-wise visualizations highlighting the total number of deaths. This allows for a comparative assessment of the mortality impact, aiding in identifying regions with higher fatality rates. Understanding the geographical distribution of fatalities is vital in formulating effective strategies. By identifying high-risk areas, authorities can implement targeted interventions, allocate resources where they are most needed, and strengthen healthcare infrastructure to better respond to the challenges posed by the virus. Fig. 4 shows the total number of deaths in each state of India due to COVID-19, with darker colors indicating more deaths. Maharashtra has the highest number of COVID-19 deaths, followed by Karnataka, Tamil Nadu, Kerala, and Delhi. These five states account for over half of all COVID-19 deaths in India. Several other states have

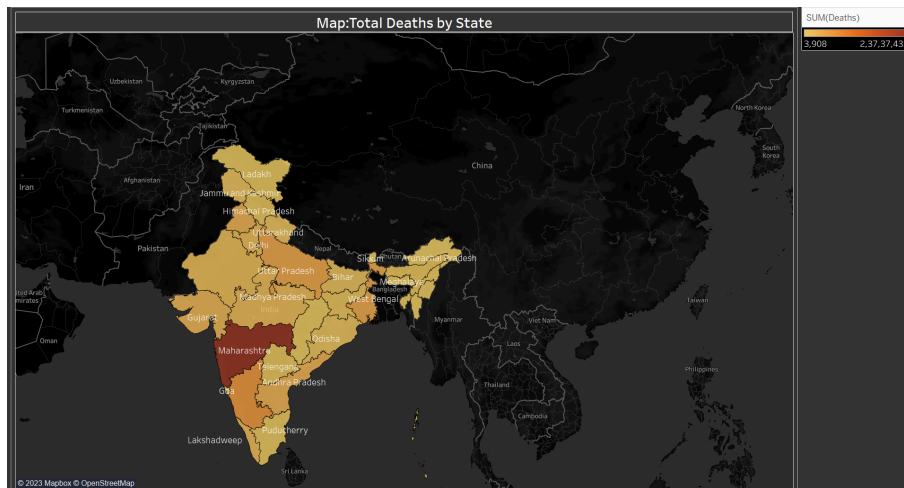


Fig. 4. Total number of deaths per state.

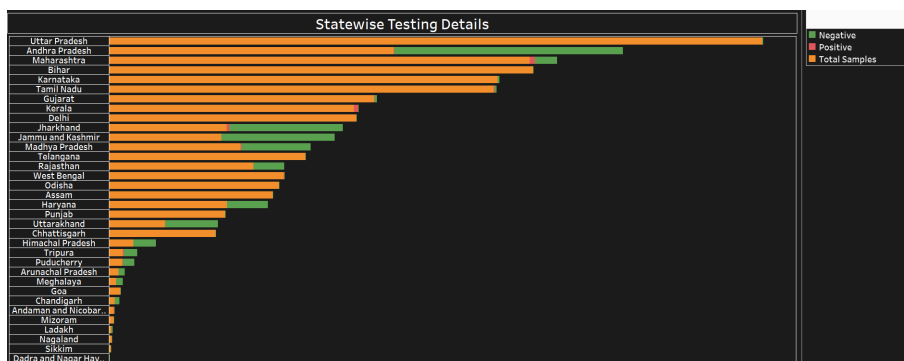


Fig. 5. Number of tests conducted by states.

also been severely affected by COVID-19, including Gujarat, Uttar Pradesh, West Bengal, Andhra Pradesh, and Madhya Pradesh. These states have each recorded over 10,000 COVID-19 deaths. The northeastern states of India have been relatively less affected by COVID-19. The only northeastern state to record over 1,000 COVID-19 deaths is Assam.

Turning attention to testing efforts, our analysis includes visualizations illustrating the number of tests conducted by different states. This offers a comparative study of testing rates and helps identify regions where testing may need to be intensified. The state-wise testing details provide a comprehensive view of our surveillance efforts against the backdrop of the pandemic. The graph shown in Fig. 5 provides raw numbers and presents how proactive each state has been in detecting and managing COVID-19. States with consistently high testing rates may effectively identify cases early, contributing to better containment strategies. States with high testing rates but low positivity rates may effectively screen and prevent outbreaks. In contrast, states with high positivity rates may need additional support to manage and control the spread. In summary, this graph presents numbers and invites us to decipher the testing strategies employed by different states. These insights are crucial for public health officials and policymakers to refine

their approaches, ensuring a proactive and effective response to the dynamic nature of the pandemic. In Fig. 5, the orange bar shows the total number of tests, whereas the green bar shows the proportion of negative cases. The higher the proportion of negative cases concerning the total number of tests, the lower the positivity rates of that state. The bar chart indicates that Uttar Pradesh is the state that has done the most tests, followed by Andhra Pradesh and Maharashtra. This large number of tests can be connected to the large population of these states.

In collaboration with the ICMR, our analysis also presents visualizations depicting testing details. This includes insights into testing methodologies and an overall understanding of the testing landscape. Shifting our attention to infrastructure, the bar graph in Fig. 6 illustrates the number of ICMR-approved testing laboratories for COVID-19 in each state. A robust network of testing facilities is vital for efficient and widespread testing. From the graph, it is clear that Maharashtra has the most significant number of ICMR-approved testing laboratories. This can also be linked to the large number of tests done in the state.

The vaccination campaign is a crucial aspect of the analysis, with visualizations providing a comprehensive overview of doses administered by the vaccine. These visuals track the

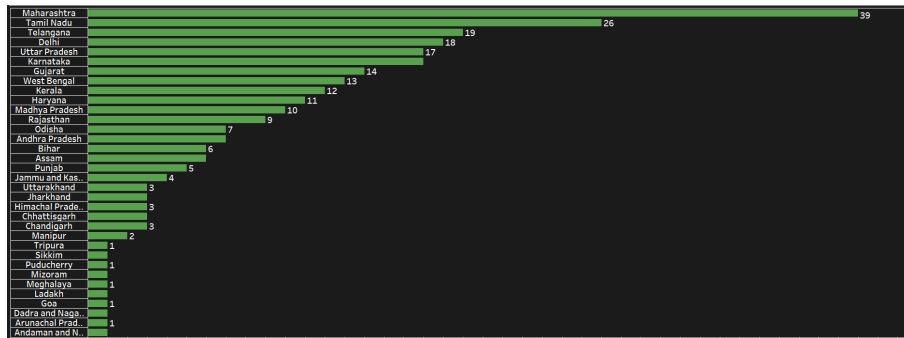


Fig. 6. Total number of ICMR testing laboratories in each state.

progress of the vaccination drive, aiding in assessing its effectiveness. The bar graph in Fig. 7 shows the number of doses administered by the vaccine in India. CoviShield is the most widely administered vaccine in India, followed by Covaxin. Sputnik V is a distant third. This is likely due to several factors, including the availability of each vaccine, its efficacy, and the cost of the vaccine.

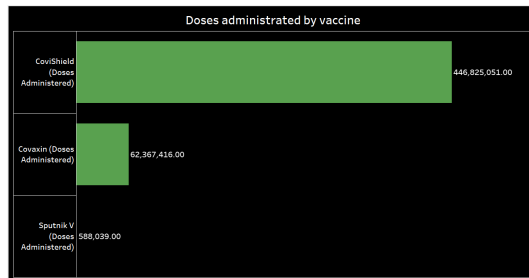


Fig. 7. Total doses of each vaccine.

Finally, a detailed examination of first versus second doses is presented through visual insights into the distribution and administration of vaccines. This holistic view of the vaccination process contributes to a comprehensive understanding of India's ongoing battle against the COVID-19 pandemic. Fig. 8 shows that the number of first-dose administrations is higher than the second-dose administration. This can be due to many reasons, including lack of awareness among people, the hesitancy of people to get the second dose of the vaccine, either because they had side effects after the first dose or because they are concerned about the safety of the vaccine, difficulty of people to schedule an appointment for the second dose.

IV. COVID-19 IN INDIA TABLEAU DASHBOARD DESIGN

The interactive dashboard offers a comprehensive overview of the COVID-19 situation in India, leveraging dynamic visualizations to distill complex data into actionable insights. Fig. 9 shows the COVID-19 in India dashboard. To create the COVID-19 in India dashboard in Fig. 9, we used a tutorial explained in [10]. The following features are highlighted from the COVID-19 in India dashboard:

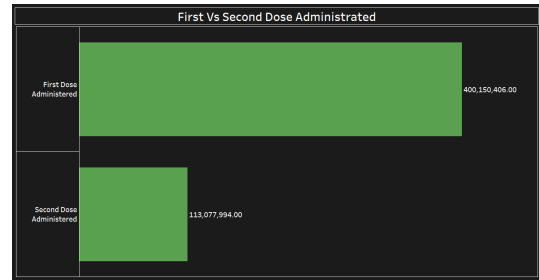


Fig. 8. Number of first Vs second dose.

- **Total Cases Overview:** Visualize the trajectory of the pandemic with real-time updates on the total number of confirmed cases across different time periods. This dynamic presentation provides a quick and intuitive grasp of the scale and progression of COVID-19 in the country.
- **Demographic Impact:** Explore the demographic distribution of confirmed cases, revealing insights into vulnerabilities within specific age groups and gender segments. These visualizations shed light on the varying impact of the virus on different segments of the population.
- **Mortality Focus:** Dive into state-wise visualizations highlighting the total number of deaths, enabling a comparative assessment of the mortality impact. Identify regions with higher fatality rates, facilitating targeted interventions and resource allocation.
- **Testing Efforts:** Evaluate testing efforts with visualizations illustrating the number of tests conducted by different states. This comparative analysis helps identify regions where testing may need intensified, contributing to a more effective containment strategy.
- **ICMR Collaboration Details:** Gain insights into testing methodologies and the overall testing landscape in collaboration with the ICMR. Understand the intricacies of testing strategies to inform ongoing public health efforts.
- **Vaccination Progress:** Track the progress of the vaccination campaign through visualizations providing a comprehensive overview of doses administered by different vaccines. Assess the vaccination drive's effectiveness in mitigating the virus's impact.

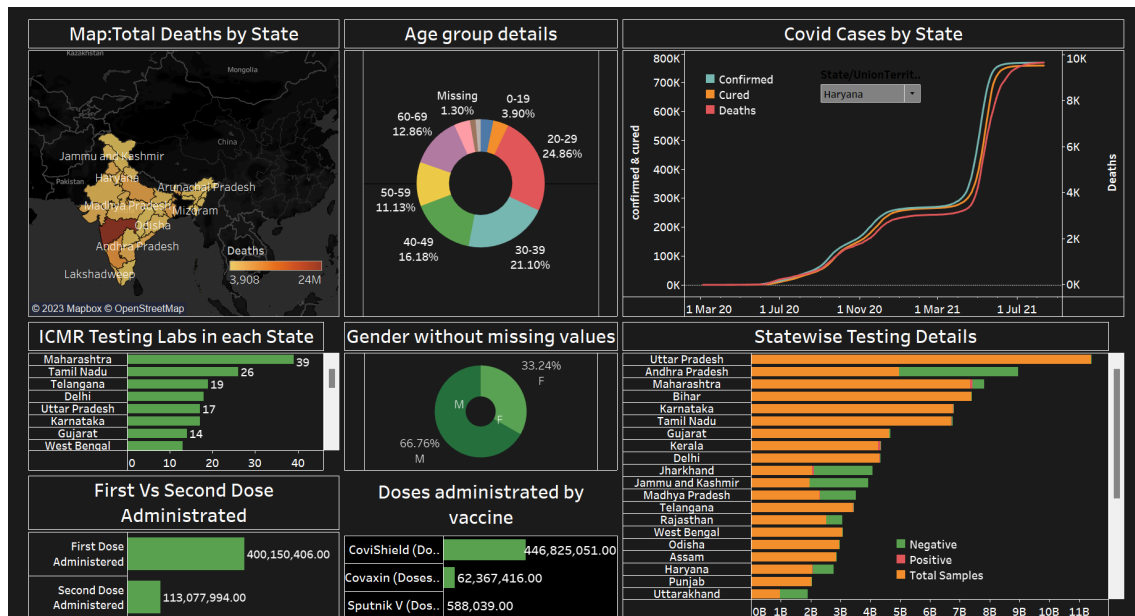


Fig. 9. COVID-19 in India dashboard.

- **First Vs. Second Doses:** Examine the distribution and administration of first and second doses, offering a detailed view of the vaccination process. This holistic perspective contributes to a comprehensive understanding of India's ongoing battle against the COVID-19 pandemic.

V. CONCLUSION

This Tableau-driven analysis has illuminated critical facets of the COVID-19 situation in India, offering nuanced insights through a comprehensive series of visualizations. The study provides a real-time and dynamic snapshot of the virus's impact by tracing the trajectory of the pandemic, from the total number of confirmed cases across distinct periods to the demographic distribution of infections. The focus on mortality rates, coupled with state-wise visualizations, enables a comparative assessment, facilitating the identification of regions facing heightened fatality rates and emphasizing the need for targeted interventions. Moreover, the scrutiny of testing efforts, including collaboration with the ICMR, not only sheds light on the current testing landscape but also underscores the importance of robust testing strategies in curbing the spread of the virus. The in-depth exploration of the vaccination campaign, delineating doses administered by vaccine and distinguishing between first and second doses, contributes valuable insights into the progress and effectiveness of the immunization drive. This holistic view equips stakeholders with actionable information for ongoing strategic decision-making in India's relentless battle against the COVID-19 pandemic. As the world navigates the evolving landscape of this global health crisis, this analysis stands as a testament to the power of data visualization in unraveling the complexities of the pandemic. We believe the insights presented herein serve as a foundation for informed policy decisions, resource

allocation, and the collective effort to safeguard public health in India and beyond.

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